Image Processing Project

Title: Intelligent Video Restoration and Enhancement System

Abstract

This project presents an intelligent video restoration and enhancement system that transforms grayscale or visually degraded videos into high-quality, fully colorized outputs. The system combines a deep learning–based colorization model with classical image and video processing techniques to restore visual information in a realistic and temporally consistent manner. Each video frame is first processed to extract luminance information, which is then passed through a convolutional neural network trained to predict plausible chrominance values. The predicted colors are refined through post-processing operations, including chroma enhancement, clipping, and spatial filtering. To ensure temporal stability across frames, a temporal smoothing strategy is applied during video reconstruction, significantly reducing flickering artifacts. The proposed approach aims to restore old footage, enhance low-quality recordings such as CCTV videos, and improve monochrome or degraded videos while preserving the original scene structure. The final output closely resembles modern film remastering pipelines, delivering visually coherent, colorized videos with enhanced clarity and realism.

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1. Introduction

1.1 Project Overview

This project presents an intelligent video restoration and enhancement system that transforms grayscale or visually degraded videos into high-quality, fully colorized outputs. The primary objective is to automatically colorize grayscale imagery by predicting plausible chrominance from luminance alone. This approach is specifically aimed at restoring old footage, enhancing low-quality recordings such as CCTV videos, and improving monochrome or degraded videos while strictly preserving the original scene structure. The final output is designed to closely resemble modern film remastering pipelines, delivering visually coherent, colorized videos with improved clarity and realism.

1.2 Technical Approach and Challenges

The core challenge in colorization lies in inferring color (chrominance) from black-and-white (luminance) data, a task often complicated by the correlation between brightness and color in standard RGB representations. To address this, the system employs a hybrid approach that integrates a deep learning–based colorization model with classical image and video-processing techniques.

The system utilizes a U-Net–based convolutional neural network (CNN) for dense pixel-level prediction. By isolating luminance from chrominance, the model must learn semantic color relationships—such as "this shape is a tree, so it should be green"—rather than merely processing brightness information. Each video frame is processed to extract luminance information, which is then fed into the network to predict plausible chrominance values.

1.3 Temporal Consistency and Enhancement

A significant challenge in extending image colorization to video is temporal instability, often manifested as flickering artifacts between frames. To ensure a stable viewing experience, this system implements a temporal smoothing strategy during video reconstruction. This involves blending chroma channels between consecutive frames to maintain smooth transitions and motion responsiveness.

Furthermore, the predicted colors undergo refinement through post-processing operations, including chroma enhancement, clipping, and spatial filtering. This ensures that the restored video not only features realistic coloration but also maintains high visual fidelity and stability, effectively bridging the gap between automated processing and professional remastering standards.

Based on the content of your document, I have organized the text into a formal university project report structure. This structure groups related concepts (Methodology, Implementation, Pipeline, and Results) to create a logical flow suitable for academic submission.

2. Methodology and System Design

2.1 Color Space Representation

The choice of color space plays a critical role in image colorization. In this work, the CIE LAB color space is adopted instead of the standard RGB representation due to its perceptual properties and its effectiveness in separating luminance from chrominance.

- Limitations of RGB: In RGB, color information is represented using three highly correlated channels that jointly encode both brightness and chromatic content, making it challenging to isolate structural details. Additionally, RGB is not perceptually uniform.
- Advantages of LAB: The LAB space explicitly decouples luminance (L) from chrominance (a, b). The L channel captures brightness and structure, while a and b represent the Green–Red and Blue–Yellow axes, respectively. This allows the model to use the L channel as a structural guide while focusing predictive power exclusively on the a and b channels.

2.2 Encoder-Decoder Architecture (U-Net)

For dense pixel-level prediction, the U-Net architecture has been selected as the standard. Its use of skip connections allows the network to preserve high-resolution spatial details from the input while simultaneously learning high-level semantic context.

This work integrates probabilistic classification logic into a U-Net–style architecture. This combination leverages the classification approach's rich color diversity while maintaining the superior spatial resolution provided by skip connections.

2.3 Proposed Approach

The present work builds upon these milestones by integrating the probabilistic classification logic of Zhang et al. into a U-Net–style architecture. This combination aims to leverage the classification approach's vibrant color diversity while maintaining the superior spatial resolution and boundary consistency provided by skip connections.

3. Problem Formulation

3.1 Input and Output Representation

Let $L \in \mathbb{R}^{H \times W}$ denote the normalized luminance channel of an image, where $H$and $W$are the image height and width. The model input is a single-channel grayscale image obtained by converting an RGB image to the LAB color space and scaling the luminance channel to the [0,1] range.

By isolating luminance from chrominance, the model is required to learn semantic color relationships rather than brightness information.

Instead of directly regressing continuous chroma values, the chrominance space is discretized into a finite number of bins. The A and B channels are each quantized into N uniform bins, resulting in $K = N^2$ discrete color classes.

Each pixel's chroma value is mapped to a class index using:

$$c = A_{bin} \times N + B_{bin}$$

This formulation transforms the colorization task into a pixel-wise multi-class classification problem, thereby enabling the model to represent uncertainty and multimodal color distributions.

3.2 Model Prediction

The model is defined as a function:

$$f_\theta : L \rightarrow P$$

where $P \in \mathbb{R}^{H \times W \times K}$represents a probability distribution over the color classes for each pixel.

A U-Net–based convolutional neural network is used to learn this mapping. The encoder extracts hierarchical semantic features from the luminance input, while the decoder reconstructs spatial resolution using skip connections to preserve fine-grained details. The final layer produces K logits per pixel, which are converted to class probabilities using the softmax function.

3.3 Loss Function and Color Prior Re-weighting

Training is performed using a pixel-wise cross-entropy loss. However, natural images exhibit a highly imbalanced color distribution, with desaturated colors appearing significantly more frequently than saturated ones. To address this imbalance, a color prior distribution is computed over the training dataset by accumulating a histogram of quantized chroma values. Let p_k denote the empirical probability of color class k. The loss is re-weighted using:

$$w_k = \frac{1}{(p_k^{\alpha} + \epsilon)}$$

The weighted cross-entropy loss is defined as:

$$\mathcal{L} = -\sum_{k=1}^{K} w_k \; y_k \log\left(p_k\right)$$

This encourages the model to predict rare yet visually salient colors, thereby improving overall color diversity and realism.

3.4 Inference and Color Reconstruction

During inference, the model outputs a probability distribution over color classes for each pixel. Instead of selecting the most likely class, a soft expectation is computed over the chroma bin centers:

$$\hat{A}, \hat{B} = \sum_{k=1}^{K} p_k \cdot (A_k, B_k)$$

Temperature scaling is applied to the SoftMax output to control color confidence. The predicted chroma maps are further refined using edge-aware bilateral filtering to reduce noise while preserving boundaries.

To enhance color vividness, a chroma gain factor is applied, followed by clipping to prevent color artifacts. The final colorized image is reconstructed by combining the predicted $a$and b channels with the original luminance channel and converting the result back to the RGB color space.

3.5 Learning Objective

The overall objective is to learn model parameters θthat minimize the expected weighted classification loss over the training dataset:

$$\theta^* = \arg\min_{\theta} \; \mathbb{E}_{(L,c)}[\mathcal{L}(f_\theta(L), c)]$$

This unified formulation captures the full colorization pipeline from grayscale input to final color reconstruction.

4. Implementation Details

This section describes the practical implementation of the project, including the curation of the training data, the automated preprocessing pipeline, and the details of data preparation for the U-Net model.

4.1 Dataset Composition

To ensure robust and semantically meaningful color prediction, the model was trained on a large-scale and diverse image dataset covering a wide range of scenes, objects, and color distributions. The dataset was carefully curated and preprocessed to maximize color quality and minimize noise that could negatively affect training. Data Sources: The training corpus was assembled from five complementary sources:

- COCO 2017: The primary source, providing complex scenes with multiple interacting objects.
- Flickr30k: Included to provide diverse real-world imagery, particularly outdoor scenes and human centered activities.
- Flickr Faces HQ (FFHQ): Integrated specifically to improve the model's accuracy in predicting realistic skin tones.
- 102 Flowers Dataset: Added to introduce highly saturated and fine-grained color examples, improving performance on botanical and natural scenes.
- Custom Internet Image Collection: Additional images were collected via public APIs to address underrepresented color distributions and scene types within the existing datasets. This multi-source strategy helped reduce dataset bias and improved the model's robustness across different visual domains.

5. Video Inference Pipeline

The pipeline processes the input video frame-by-frame while maintaining temporal coherence.

5.1 Preprocessing and Batching

The input video is decoded into individual frames, preserving the original frame rate and resolution. Frames are processed in batches to improve GPU utilization. Each frame is resized to a fixed resolution (256 * 256), converted to LAB space, and the chrominance channels are discarded, leaving only the normalized L channel.

5.2 Neural Network Inference

The batch of luminance frames is passed through the U-Net model, which outputs a probability distribution over discretized color bins. Temperature scaling is applied to control the sharpness of the predicted distributions. The probabilities are converted into continuous chroma values using a soft expectation over color bin centres.

5.3 Temporal Smoothing and Post-Processing

A temporal smoothing mechanism is applied to address flickering. The chroma channels of the current frame are blended with those of the previous frame using a temporal blending factor. Following smoothing:

1. Chroma Enhancement: A gain factor is applied to enhance color vividness.
2. Upscaling: Predicted chroma channels are upscaled to the original video resolution.
3. Refinement: Edge-aware bilateral filtering is used to reduce noise while preserving boundaries.

5.4 Video Reconstruction and Final Output

5.4.1 Overview

Video reconstruction is the final stage of the proposed pipeline, where the individually processed and colorized frames are reassembled into a continuous video sequence. The goal of this stage is to preserve the input video's original temporal structure, spatial resolution, and motion characteristics while incorporating the enhanced color and visual improvements generated during processing.

5.4.2 Frame Integration

Each processed frame is generated independently, but in the correct temporal order. During reconstruction:

- Frames are written sequentially to the output video stream.
- The original frame rate (FPS) of the input video is preserved.
- No frames are dropped or duplicated, ensuring smooth motion playback.

This approach guarantees temporal alignment between the input and output videos.

5.4.3 Preservation of Spatial Properties

The reconstructed video maintains the original frame resolution (width and height), aspect ratio, and consistent spatial alignment across frames. Since chroma information is upsampled to the original resolution before reconstruction, color details remain aligned with edges, textures, and motion boundaries.

5.4.4 Color Space Conversion for Output

Before writing each frame to the output video, it is converted from the LAB color space to RGB/BGR. This step ensures compatibility with standard video codecs and media players. By performing all processing in LAB and converting only at the final stage, color manipulation remains perceptually stable and visually accurate.

5.4.5 Final Visual Refinement

A mild spatial sharpening operation is applied immediately before reconstruction. This enhancement is intentionally subtle to avoid ringing artifacts or noise amplification. Its purposes are to:

- Enhance perceived detail.
- Improve visual clarity.
- Compensate for any softness introduced during resizing or filtering.

5.4.6 Video Encoding

The processed frames are encoded into a standard video format. To ensure the reconstructed video can be played back smoothly on common media platforms without additional processing, the system uses:

- The same frame rate as the original video.
- A widely supported video codec.
- Sequential frame writing to preserve timing accuracy.

5.4.7 Resulting Output

The final reconstructed video retains the structure and motion of the original footage while exhibiting stable, temporally consistent colorization. It shows improved visual quality and clarity, avoiding flickering, color bleeding, and temporal artifacts

6. Image and Video Processing Techniques Applied

This project applies a combination of classical image processing and video processing techniques alongside deep learning to achieve high-quality restoration[1].

6.1 Color Space Transformation

- Technique: LAB Color Space Conversion. Each video frame is converted from RGB/BGR to the CIE LAB color space.
- Why this is image processing: LAB separates luminance (L) from chrominance (a, b). Structural details, edges, and motion are preserved in the luminance channel, while color processing is applied only to chroma channels[4]. This separation enables controlled color manipulation without affecting brightness or contrast.

6.2 Spatial Resampling (Resizing)

- Technique (Downscaling): Frames are resized to a fixed resolution (256*256) before processing to reduce computational cost and standardize input dimensions.
- Technique (Upscaling): Predicted chroma channels are resized back to the original video resolution using high-quality interpolation[7].
- Image processing role: This preserves spatial alignment and avoids blocky artifacts in the final output.

6.3 Luminance Preservation

- Technique: L Channel Locking. The original luminance channel is never modified during processing.
- Why this matters: This preserves edges, texture, motion, and lighting while preventing brightness flickering. It ensures temporal consistency and is a classic luminance–chrominance processing strategy[11].

6.4 Chroma Estimation and Enhancement

- Technique: Chroma Amplification. After prediction, the chroma channels are scaled to increase color visibility.
- Image processing role: Enhances color vividness and compensates for conservative model predictions.
- Technique: Value Clipping. Chroma values are constrained to a valid range to prevent color overflow and avoid unrealistic or unstable colors.

6.5 Temporal Filtering (Video Processing)

- Technique: Temporal Smoothing. Each frame's chroma channels are blended with those of the previous frame.
- Why this is video processing: This is equivalent to a temporal low-pass filter applied across time. It reduces flickering, stabilizes color transitions, and maintains smooth motion. This is a purely video-processing technique, independent of deep learning.

6.6 Noise Reduction and Artifact Control

- Technique: Implicit Noise Suppression. The system employs a noise-reduction pipeline for chroma information, where batch inference smooths prediction noise, temporal blending suppresses random fluctuations, and clipping removes extreme outliers.

6.7 Spatial Enhancement (Sharpening)

- Technique: Mild Unsharp Masking. A light sharpening filter is applied after color reconstruction.
- Purpose: This classic spatial enhancement technique enhances perceived detail and improves visual clarity without introducing over-sharpening artifacts.

6.8 Video Reconstruction

- Technique: Frame Reassembly. Processed frames are written back into a video stream while preserving the original resolution, frame rate, and temporal order.
- Purpose: This ensures the output video remains visually coherent and synchronized.

7. Results and Discussion

7.1 Qualitative Results

The final reconstructed video successfully retains the structure and motion of the original footage. The output exhibits stable, temporally consistent colorization and shows improved visual quality and clarity. Notably, the system effectively avoids common restoration issues such as flickering, color bleeding, and temporal artifacts.

7.2 Performance Analysis

The overall performance of the proposed system was evaluated as good. The model demonstrated the ability to predict correct colors for most objects in the video sequences. It also exhibited strong edge awareness, resulting in minimal color bleeding between adjacent

regions. The colors produced were vibrant and visually appealing, contributing to a lively, realistic appearance in the reconstructed videos.

7.3 Limitations and Future Improvements

Despite its success, the system exhibited limitations in handling complex lighting conditions. In certain instances, the model exhibited overconfidence in predicting chroma values across a wide a-b color range, which occasionally resulted in unnatural color intensities or localized color artifacts.

These limitations are primarily attributable to the scale and diversity of the training data and the training duration. To address this, future iterations of the model would benefit from:

- Expanded Dataset: Training on a significantly larger dataset (approximately 1–1.5 million images).
- Extended Training: Increasing the training process to 150–250 epochs.

It is expected that these adjustments would improve color confidence calibration, lighting robustness, and overall generalization performance.

References

- Zhang, R., Isola, P., & Efros, A. A. (2016). Colorful Image Colorization. *Proceedings of the European Conference on Computer Vision (ECCV)*.

- Ronneberger, O., Fischer, P., & Brox, T. (2015). U-Net: Convolutional Networks for Biomedical Image Segmentation. *Proceedings of the International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI)*.

- Cheng, Z., Wang, L., & Liang, Y. (2017). Colorization using Deep Convolutional Neural Networks. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*.

- Larsson, G., Matusik, W., & Felsberg, M. (2017). Learning Representations for Automatic Colorization. *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*.

- Fairchild, M. D. (2005). *Color Appearance Models* (2nd ed.). Addison-Wesley.

- Tomasi, C., & Manduchi, R. (1998). Bilateral Filtering for Gray and Color Images. *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*.

- Isola, P., Zhu, J. Y., Zhou, T., & Efros, A. A. (2017). Image-to-Image Translation with Conditional Adversarial Networks. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*.

- Iizuka, S., Simo-Serra, E., & Ishikawa, H. (2016). Let There Be Color: Joint End-to-End Learning of Global and Local Image Priors for Automatic Image Colorization with Deep Learning. *ACM SIGGRAPH Conference on Computer Graphics and Interactive Techniques*.

- Goodfellow, I., Pouget-Abadie, J., Mirza, M., Xu, B., Warde-Farley, D., Ozair, S., Courville, A., & Bengio, Y. (2014). Generative Adversarial Nets. *Neural Information Processing Systems (NeurIPS)*.

- Long, J., Shelhamer, E., & Darrell, T. (2015). Fully Convolutional Networks for Semantic Segmentation. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*.

- Iizuka, S., Simo-Serra, E., & Ishikawa, H. (2016). Let There Be Color: Joint End-to-End Learning of Global and Local Image Priors for Automatic Image Colorization with Deep Learning. *ACM SIGGRAPH 2016*.

- Perez, L., & Wang, J. (2017). The Effectiveness of Data Augmentation in Image Classification Using Deep Learning. *CVPR Workshop — Convolutional Neural Networks for Visual Recognition*.

- Buda, M., Maki, A., & Mazurowski, M. A. (2018). A Survey of Deep Learning with Data Augmentation in Medical Imaging. *Journal of Medical Imaging, 5(2), 1–17*.

- Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. A., Kaiser, Ł., & Polosukhin, I. (2017). Attention is All You Need. *Neural Information Processing Systems (NeurIPS)*.

- Gatys, L. A., Ecker, A. S., & Bethge, A. (2015). A Neural Algorithm of Artistic Style. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*.

- Choi, Y., Choi, M., Ha, J., & Kim, J. (2018). Image Colorization with Generative Adversarial Networks. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*.

- Johnson, J., Alahi, A., & Fei-Fei, L. (2016). Perceptual Losses for Real-Time Style Transfer and Super-Resolution. *Proceedings of the European Conference on Computer Vision (ECCV)*.

- Donahue, J., Jia, Y., Vinyals, O., Hoffman, M., & Darrell, T. (2016). DeCAF: A Deep Convolutional Activation Feature for Generic Visual Recognition. *Proceedings of the International Conference on Machine Learning (ICML)*.